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Review

# Deciphering and harnessing gut microbiota–associated immune regulation in acute graft-versus-host disease

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Allogeneic hematopoietic stem cell transplantation represents a curative treatment of choice for numerous severe hematological malignancies. While donor-derived transplanted T cells can limit disease relapse (GvT/GvL effect), they also induce, in 30–50% of the patients, acute graft-versus-host disease (aGvHD), a severe condition with elevated mortality and comorbidity rates. Gut microbiota composition has been associated with aGvHD outcome. This observation created a substantial research interest, and individual gut microbiota commensals have been acknowledged for their ability to promote immune regulation, both locally and systemically, and thus limit aGvHD-related inflammation. The mechanisms by which commensals support immune homeostasis are being decrypted at a remarkable rate. However, the trillions of micro-organisms comprising the gut microbiome interact, both directly and indirectly, with local immune cells, which is all the more critical in the context of heavy conditioning regimens these patients undergo, themselves damaging mucosal tissues and prompting inflammation. Commensals can help preserve the gut barrier integrity by actively limiting deleterious inflammation processes. Mechanistically deciphering the intricate crosstalk between gut microbes and gut immune cells, both at the species level and globally, represents a colossal challenge, but holds great promise in predicting and harnessing numerous pathological processes, including aGvHD. This review aims to examine the acquired knowledge concerning immunoregulatory responses driven by gut microbiota in the context of aGvHD. Recent preclinical and clinical studies harnessing such pathways proved to be encouraging, while substantial hurdles subsist regarding how to successfully harness this complex host/microbiota interplay to constrain aGvHD.

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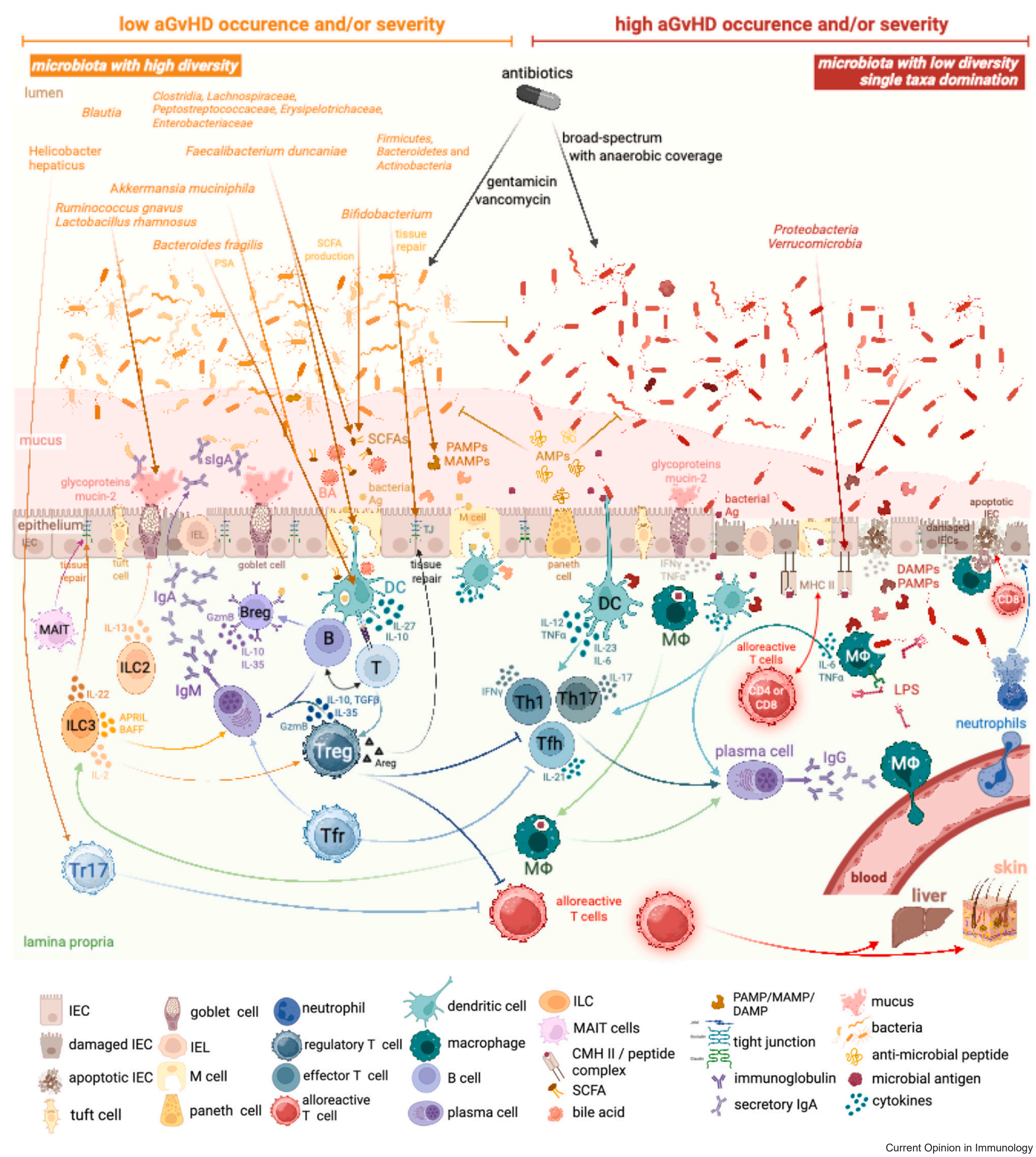
## Introduction

Patients undergoing severe hematological disorders, mainly malignant, are commonly treated with allogeneic hematopoietic stem cell transplantation (allo-HSCT). This treatment consists of the administration of myeloablative or reduced-intensity conditioning regimens to eliminate aberrant immune cells, followed by the infusion of donor-derived hematopoietic stem cells to reconstitute the hematological and immune systems in the patients. Graft-versus-host disease (GvHD) represents a frequent complication ensuing allo-HSCT with high morbidity and mortality [1] and is elicited when transplanted donor-derived immune cells attack host tissues and organs [2,3]. Acute GvHD (aGvHD) occurs in 30–50% of allo-HSCT–treated patients and predominantly affects the gastrointestinal tract, the skin, and/or the liver, essentially within 100 days post-transplantation, while chronic GvHD (cGvHD) typically appears later on and can affect multiple organs. Mechanistically, the conditioning treatment can first damage patients’ tissues, prompting inflammation and subsequent activation of antigen-presenting cells (APCs), which in turn can activate donor-derived alloreactive proinflammatory $CD4^+$ (e.g. Th1, Th17) and cytotoxic $CD8^+$ T cells (e.g. Tc1) that can then migrate into host tissues. The resulting overall immune activation, cell death, and tissue damage also promote the release of Pathogen-Associated Molecular Patterns (PAMPs), Microbe-Associated Molecular Patterns (MAMPs), and Damage-Associated Molecular Patterns (DAMPs), which can further fuel inflammation and dictate aGvHD symptoms and severity (Figure 1).

**Figure 1**




The 'gut microbiota/immunoregulatory responses' interplay in aGvHD onset and severity. High microbiota diversity and the presence of key commensals protect against aGvHD through priming, activating, and/or modulating immunoregulatory cell subsets, which in turn promote gut barrier integrity. On the other hand, low gut microbiota diversity as well as single taxa domination can foster proinflammatory cues and alloreactive T cell activation and migration, thereby compromising gut barrier integrity and fueling inflammation, promoting and dictating aGvHD symptoms. Inflammation can in return disrupt microbiota resilience and lead to long-lasting alterations of the host–microbe balanced relationship. Created in BioRender. Godefroy (2025) https://BioRender.com/3bm1rxm.

Steroids represent the first-line treatment for aGvHD. However, approximately 50% of aGvHD patients become resistant. Ruxolitinib was approved by the Food and Drug Administration (FDA) in 2019 as a second-line treatment for steroid-refractory or -dependent aGvHD, but over 60% of these patients do not respond to Ruxolitinib [4]. Therefore, the lack of predictive biomarkers and novel therapeutic avenues constitute a major unmet medical need.

Over the last decade, intensive ongoing research has been revealing the colossal impact of gut microbiota on numerous physiological processes and diseases' susceptibility. The symbiotic relationship between gut microbes and the host organism is essential for the latter to maintain tissue homeostasis and sustain general health, far beyond the intestinal tract. Both qualitative and quantitative disruptions of this complex ecosystem, so-called dysbiosis, have been shown to be implicated in a large array of inflammatory conditions, including aGvHD [5]. The gut microbiome mainly comprises four bacterial phyla: *Firmicutes* (or *Bacillota*) and *Bacteroidetes*, constituting 90% of gut bacteria, as well as *Actinobacteria* and *Proteobacteria* [6]. The abundant *Bacteroides* genus is particularly important for the production of short-chain fatty acids (SCFAs) from the breakdown of complex dietary carbohydrates. Of note, among SCFAs, butyrate stands out as it is significantly involved in several key processes such as gut barrier integrity maintenance [7–9], immunoregulation [10–12], or metabolism [13]. Among *Firmicutes*, *Faecalibacterium*, *Roseburia*, and *Bifidobacterium* also represent common, abundant genera potently contributing to gut health, immunoregulation, and metabolic functions [14] (Figure 1). In addition to bacteria, gut microbiota also contains archaea, eukaryotes, for example, yeasts, and a diverse array of viruses, predominantly phages [15]. This review will focus on commensal bacteria and their interplay with immune-regulatory responses relevant to aGvHD incidence and outcome.

## Gut microbiota impact on aGvHD

### Gut barrier

Intestinal epithelial cells (IECs) have evolved the ability to prevent the translocation of potentially harmful microbiota components or dietary products from the gut lumen into the body, mainly through the formation of tight junctions, relying on a protein network (e.g. occluding, claudin, zonula occludens) to preclude gut permeability. Different bacteria can modulate the expression of such proteins (Figure 1). For instance, several *Bifidobacterium* strains can upregulate tight junctions, β-catenin, claudin-3, occludin, and ZO-1, along with increased barrier-modifying mucin 2, IL-10, and PPARγ, while downregulating proinflammatory TNFα and IL-6 [16]. In addition, IECs exert multiple immune functions upon microbiota exposure through their expression of Pattern Recognition Receptors (PRRs), including cytokines (e.g. IFNγ, TNFα, TGFβ) to coordinate immune responses, or APRIL, involved in B cell differentiation and IgA class-switching [17]. Furthermore, IEC types include goblet cells that produce mucin glycoproteins forming a protective mucus layer, which can partly be stimulated by *Ruminococcus gnavus* [18] and *Lactobacillus rhamnosus* [19], and protecting against invading microbes [20]; Paneth cells at the bottom of intestinal crypts that secrete antimicrobial peptides (AMPs; e.g. RegIII and lipocalin-2), microfold (M) cells involved in antigen capture and presentation, and tuft cells that promote type-2 immunity to intestinal parasites [17,21] (Figure 1).

Upon conditioning treatments, inflammation, and cell death, these barriers can be compromised through tight junction disruption and lead to the translocation of gut microbes and their derivatives, such as PAMPs/MAMPs, bacterial antigens recognized by T and B lymphocytes, proteins/peptides with molecular mimicry features as well as metabolites, which can then reach and shape local immune responses [21] (Figure 1). Inflammation can in turn disrupt microbiota resilience, that is, lead to long-lasting alterations of the host–microbe balanced relationship due to this vicious circle [22]. The observation that most epithelial barriers for organ-specific microbiota ecosystems actually correspond to aGvHD target organs supports the idea that microbiota composition represents an early key player in aGvHD development and severity.

### Gut microbiota composition affects aGvHD onset and outcome

Gut microbiota represents a complex network, with trillions of micro-organisms, comprising over 1000 bacterial species, providing a myriad of metabolic and immunological functions benefiting the host as well as its own resilience/stability partly through competitive exclusion of potential pathogens [14,15,23]. As opposed to domination with single taxa [24], peri-transplant gut microbiota diversity is consistently associated with health and is believed to lessen GvHD risk and severity [5,25–27] (Figure 1). The types of antibiotics used also appear to be critical. For instance, patients who receive broad-spectrum antibiotics with anaerobic coverage could be at increased risk for severe aGVHD, particularly when antibiotics are administered peri-transplant [28–30]. A mouse study utilizing cohousing and antibiotic treatments demonstrated that distinct bacteria, again selected through different types of antibiotics, could either promote or control GvHD through inducing or inhibiting class II MHC molecule expression on IECs, independently of donor–host genetic disparity [31] (Figure 1). More precisely, a large proportion of bacterial MHC-II inducers were vancomycin-sensitive, and peri-transplant oral vancomycin administration attenuated $CD4^+$ T cell–mediated GvHD [31]. Illustrating this, a decrease in the abundance of beneficial bacteria, including *Firmicutes* (particularly, *Clostridium*, *Faecalibacterium*, *Lachnospiraceae*, *Ruminococcaceae*, *Eubacteriaceae*, and *Peptostreptococcaceae*), *Bacteroidetes* (*Bacteroides* and *Parabacteroides*), and *Actinobacteria*, has been linked to increased GvHD risk and severity [32]. In contrast, an increase in *Proteobacteria* (especially *Gamma-Proteobacteria* and *Enterobacteriales*) or *Verrucomicrobia* has been associated with worse aGvHD outcome [33,34]. These data underscore the imperative need for, on the one hand, careful consideration of antibiotic use in allo-HSCT patients and, on the other hand, a greater in-depth understanding of underlying mechanisms used by individual microbes promoting immunoregulation and ultimately protecting against aGvHD (Figure 1). These represent a substantial research interest and hold great promise, as exemplified by the first-in-human multicentric phase IIa trial and a compassionate expanded access program using a diverse fecal microbiota product, so-called MaaT013, for the treatment of refractory gastrointestinal (GI) aGvHD, which achieved outstanding overall response rates of 38% and 58%, respectively [35].

## THE gut microbiota fine-tunes immune responses and regulation, significantly weighing on aGvHD onset and outcome

The myriad of gut colonizing bacteria delivers abundant yet distinct or partially overlapping immunoregulation cues, which need to be deconvoluted, ideally at the strain-level resolution, and at a mechanistic level, to envisage relevant therapeutic strategies that could modulate gut microbiota–related immunity to improve aGvHD outcome, among other conditions. In line with this, microbiota can control immune responses, comprising pathogenic alloreactive T cells, through various avenues [36], such as APC modulation (e.g. induction of tolerogenic dendritic cells [DCs] or limiting expression of class II molecules), regulatory T cell induction/activation, providing microbe-derived antigens recognized by regulatory or homeostatic T and B cells or producing metabolites with regulatory functions (e.g. SCFAs, including butyrate, or bile acids) (Table 1) (Figure 1). Of note, in this context, regulatory cells' influence on the GvT/GvL effect, and ultimately disease relapse, needs to be carefully assessed.

### *Cells of myeloid origin*

Myeloid cells comprise innate immune cells including monocytes, macrophages, and DCs. These APCs are differentially activated by gut microbiota toward a continuum from pro- to anti-inflammatory profiles. They are key to prime, activate, and skew functionally plastic immune responses.

Mouse models suggest that monocyte-derived DCs contribute to aGVHD triggering [37]. Also, the level of nonclassical monocytes in bone marrow grafts can represent risk factors for later aGVHD [38]. The role of monocytic myeloid-derived suppressor cells in G-CSF-mobilized peripheral blood grafts remains elusive, and these cells have been reported either as aGvHD risk factor [39] or potentially as protective through their nonspecific suppression of several immune cells [36]. Several microbiota-derived bacteria, such as certain *Bifidobacterium* strains [16], *Bacteroides fragilis* (via PSA) [40], or *Faecalibacterium duncaniae* (previously named *F. prausnitzii*) [41], have been shown to promote tolerogenic DCs and/or macrophages and subsequent Treg induction (Figure 1) through diverse mechanisms, including NF-κB inhibition and ensuing proinflammatory cytokine inhibition, anti-inflammatory cytokine production, TLR2/6 and TLR5 triggering, dampening of their co-stimulation and antigen-presentation properties, or both direct and indirect (e.g. through cross-feeding mechanisms) SCFAs (derived from host extrinsic sources, such as diet and microbiome, or from host intrinsic sources that are further processed or used by the gut microbiome) and secondary bile acid production [8,16,40–42] (see Table 1 for a more comprehensive list).

Macrophages are APCs displaying great heterogeneity and plasticity, thus harboring distinct phenotypes and functions depending on tissue-specific microenvironmental cues. They are distinguished according to different classification methods, with inconsistent markers, impeding research investigations [43]. Nonetheless, donor-derived CD11c$^+$/CD14$^+$ macrophages dominate in skin GVHD, exhibiting a heightened ability to activate allogeneic T cells, secrete high levels of proinflammatory cytokines, and mediate skin damage [44]. Host macrophages are also involved in aGvHD by responding to tissue damage and immune stimuli, promoting inflammation through the recognition of DAMPs and PAMPs/MAMPs, and subsequent lymphocyte stimulation. Accordingly, inhibiting macrophage recruitment has been shown to alleviate GvHD [45]. Macrophages that infiltrate aGvHD-associated tissues usually display proinflammatory features and are commonly named M1 macrophages. In contrast, M2 macrophages, producing anti-inflammatory cytokines such as IL-10 and TGFβ, are usually associated with anti-inflammatory and wound healing properties (e.g. as a source of Areg), unexpectedly predominate in refractory aGvHD [46]. Although they can mitigate tissue damage and promote recovery, an excessive or dysregulated M2 response may also contribute to GVHD, particularly cGvHD, by supporting a fibrotic environment in affected organs. Specific gut microbiota bacteria can affect APCs' profile. For instance, *Akkermansia muciniphila* mitigates inflammation through multiple mechanisms, including the adenosine monophosphate–activated protein kinase pathway activation via TLR-2 stimulation and NLRP3 inflammasome activation [47]. Further work is needed to clarify

**Table 1**

**Gut microbiota's beneficial impacts on aGvHD onset and related complications.**

| Bacteria | Association with aGVHD | Demonstrated or suspected mechanisms | Refs. |
|---|---|---|---|
| *Akkermansia* genus | reduced GvHD risk | SCFA production, increased Treg induction | [29,105] |
| *Akkermansia muciniphila* (at least, certain strains) | reduced aGvHD risk | SCFA production, increased Treg induction | [106] |
| *Bacillus spp.* | reduced aGI-GvHD risk | decreased MyD88/TLR9 triggering | [107] |
| *Bacteroides fragilis* | reduced both aGvHD (and cGvHD) | glycoantigens (PSA) production, inducing human peripheral Tr1 cell differentiation | [108,109] |
| *Bacteroides ovatus* | reduced GvHD severity, reduced steroid-resistant aGI-GvHD, reduced GvHD-related death | xylose production, suppression of mucus-degrading gut microbes | [110] |
| *Bifidobacterium* | reduced aGvHD risk | increased production of metabolites, i.e., polyamines (N-acetyl putrescine and N-acetyl spermidine), secondary bile acids (isoDCA), | [40] |
| *Blautia* genus | reduced GvHD-related death, reduced GvHD severity, reduced aGvHD risk | increased Treg/Th17 ratio, SCFA production (butyrate, which binds to GPR43) | [105,106,111–114] |
| Clostridiales order (*Faecalibacterium, Roseburia, Ruminococcus, Blautia*) | reduced GvHD risk | Specific Foxp3+ Treg induction in mouse colon; SCFA production | [107,115] |
| *Clostridium butyricum* | reduced aGvHD risk | *maintenance of intestinal microbiota diversity* | [116] |
| *Coriobacteriaceae* family | reduced aGI-GvHD risk | inflammation prevention, possibly in a TLR4/LPS-dependent manner | [117] |
| *Faecalibacterium duncaniae* | reduced GvHD risk | TolDC induction, gut microbiota-reactive DP8α Treg induction, NF-κB inhibition via MAM, SCFA (particularly potent butyrate producer) production | [41,65,118–121] |
| *Faecalibacterium* genus | reduced GvHD risk | SCFA production, increased Treg induction | [105] |
| *Lachnospiraceae* family | reduced aGvHD risk | SCFA production, increased Treg/Th17 ratio | [105,114,118] |
| *Lactobacillus* | reduced GvHD risk | SCFA (mainly, lactate) production, inflammation, and gut damage prevention | [114] |
| *Ruminococcaceae* genus | reduced aGvHD risk | SCFA production, increased Treg induction | [105,114,118] |
| *Veillonella* genus | reduced aGvHD risk | SCFA (propionate) production | [105,118] |
| lactase-producing bacteria (mainly, *Actinobacteria, Proteobacteria, Firmicutes*) | reduced GvHD risk | SCFA (mainly, lactate) production, inflammation prevention, and increased Treg induction | [114,122] |
| gentamicin-resistant bacteria | reduced GvHD-related death | modulation of MHC-II expression on IECs | [31] |
| vancomycin-resistant bacteria | reduced GvHD-related death | modulation of MHC-II expression on IECs | [31] |

Indicated gut microbiota bacteria are reported to protect against aGvHD. Their demonstrated or suspected underlying mechanisms are also indicated.

whether heavy macrophage infiltration, especially CX3CR1+ gut macrophages that, similarly to CD103+ DCs, can sample bacterial antigens from the lumen, exacerbates or attenuates the severity of GvHD in humans [48].

### *Regulatory T cells*

Tregs have been considered a key promising cell type to target in aGvHD, as they can both curtail inflammation and promote tissue repair.

Tregs can be induced by certain gut bacteria. For instance, clusters IV and XIVa of the genus *Clostridium* promoted specific RORγt-expressing Foxp3+ Treg accumulation in the mouse gut, thereby alleviating colitis [12,49]. As mentioned above, *Bacteroides fragilis* [40] and *Faecalibacterium duncaniae* [41] have been shown to promote tolerogenic APCs and subsequent Treg induction in mice and humans, respectively. *Helicobacter hepaticus* also induces specific regulatory/homeostatic Th17 in a c-MAF-dependent manner [50]. Underlying mechanisms involve SCFA production and APC tolerization (Figure 1).

Human CD25$^{HIGH}$/CD127$^{LOW}$ Foxp3+ Tregs have long been contemplated for cellular-based therapy against aGvHD. Unlike mouse RORγ+/Foxp3+ Tregs, human Foxp3+ Tregs do not seem prone to respond to gut microbiota in a TCR-dependent manner, but rather through bystander effects. Low Foxp3+ Treg frequency, along with deficiency in Treg reconstitution and Treg function, is associated with aGvHD in preclinical and clinical studies, likely through unleashing alloreactive T cell activation [51–53]. Conversely, prophylactic adoptive transfer of donor-derived Foxp3+ Tregs can mitigate aGvHD [52–54]. However, their use in cellular-based therapy remains challenging, mainly due to their limited proliferation capacity, instability (e.g. reverting into Treg 'wannabes'), and lack of specific markers in humans to distinguish them from activated effector T cells also transiently expressing Foxp3 and activation marker, including CD25 [55,56].

Type 1 Tregs (Tr1) do not express Foxp3 and are induced in the presence of high levels of IL-10, possibly coming from microbiota-exposed anti-inflammatory APCs, and IL-27 [57]. They were first detected in tolerant allo-HSCT patients [58]. In mouse models, Tr1 cell deficiency has been associated to exacerbated aGvHD [59] and have demonstrated their ability to control GvHD [60]. Their potential clinical effects on GvHD should unravel as enhanced methods to produce pure and stable Tr1 cells arise, and specific, reliable markers are identified [36].

A distinct Foxp3-negative human Treg subset, named DP8α Tregs, was identified a few years ago [61]. DP8α Tregs are reactive to gut microbiota *Faecalibacterium duncaniae/prausnitzii*, display a unique $CD3^+/CD4^+/CD8\alpha^{LOW}/CD25^{INT}/CCR6^+/CXCR6^+$ phenotype and potent pleiotropic suppressive functions on T, B, and myeloid cells, mainly driven by CD39, CD73, and IL-10 [41,61–63]. They are abundant in the gut lamina propria and recirculate in the blood [61,62]. DP8α Tregs hold great promise in several inflammatory diseases [61,62,64], including aGvHD [65]. Indeed, low levels of circulating $CD73^+$ DP8α Treg one-month post-allo-HSCT were associated with aGvHD occurrence, without any significant impact on hematological disease's relapse. They are particularly prone to cross-react to allo-antigens [65], and proved particularly relevant in aGvHD as low amounts of host-reactive DP8α Tregs associated with aGvHD [65]. Importantly, human DP8α Tregs dramatically protected mice against xeno-GvHD through limiting deleterious inflammation and preserving gut integrity [65]. Altogether, these results demonstrate that human DP8α Tregs prevent aGvHD, likely through host-reactivity, advocating for their use in cellular-based therapy to preclude aGvHD-related inflammation. Moreover, from a clinical standpoint, DP8α Tregs harbor remarkable proliferative potential, stability, chemotactic predilections for inflamed tissues, and TCR specificities against *F. duncaniae* (commensal decreased in aGvHD patients; see Table 1) and allo-antigens, characteristics particularly suited and relevant to prevent aGvHD. Further, once infused, DP8α Tregs could be restimulated through probiotics or microbiota-derived antigens/epitopes administration to sustain long-lasting therapeutic efficacy, as demonstrated in mice [64].

Other Treg subsets could potentially contribute to aGvHD protection, including $CD8^+$ Tregs [66,67], Tfr (the Tfh/Tfr ratio can be modulated by *Ruminococcus* [68], through their crosstalk with B cells) [69,70] or commensal-elicited homeostatic, although plastic, Th17 cells [71,72]. However, these Treg subsets generally remain ill-defined and need further characterization.

It is also worth mentioning that Treg cells could also be engineered for CAR-mediated immunosuppression, protecting against xeno-GvHD in NSG mice [73].

### Regulatory B cells

Regulatory B cells (Bregs) have been investigated for their ability to curb inflammation and inhibit effector T cells in transplantation settings. Bregs come in different flavors, making their study challenging. Indeed, Bregs present heterogeneously described markers, have limited *in vitro* expansion capacities, and their purification is yet to be perfected. As a consequence, $IL\text{-}10^+$ Bregs displaying different phenotypes have been described, including $CD24^{HIGH}/CD27^+$, $CD24^{HIGH}/CD38^{HIGH}$, $CD27^{INT}/CD38^+$, or inducible $CD73^-/CD25^+/CD71^+$, and their state, differentiation, and functions depend on their environment [74,75].

$IL\text{-}10^+$ Bregs can be induced through various stimulatory cocktails of TLR agonists, BCR engagement, CD40 ligation, and/or various recombinant cytokines, as well as microbiota components, including *Clostridia* and SCFAs [76] (Figure 1). Recent studies reported different or overlapping $IL\text{-}10^+$ Breg-containing populations. For instance, induced CD1c-expressing Bregs, containing ≈30% of $IL\text{-}10^+$ Bregs, successfully alleviated GvHD in a humanized mouse model [77]. Likewise, IL-35-producing Bregs [78] mitigated murine aGvHD through T cell inhibition/anergy/exhaustion by inducing immune checkpoint (LAG-3, PD-1, or CTLA-4) expression on alloreactive and bystander T cells, in a unique IL-35-dependent manner [79]. While, serum levels of Granzyme B (GzmB) were associated with aGvHD incidence and severity [80], GzmB-producing Bregs can convey regulatory functions [81], be found in tolerant renal transplantation patients [82], and be induced in the presence of IL-21 and phorbol-12-myristate-13-acetate, but were rather unanticipatedly correlated with both occurrence and severity of GvHD, although in the chronic state of the disease [83].

One of the mechanisms that has evolved to maintain microbiota/host symbiosis and gut homeostasis relies on B cell production of IgA, coating commensals, thereby preventing their interaction with IECs and fostering a diverse, healthy microbiota (Figure 1). Interestingly, total and flagellin-specific $IgA^+$ B cells, induced by flagellin-specific Tregs in mice [84], could also participate in limiting aGvHD development, particularly in the gut. On the other hand, commensal-reactive IgA has also been reported to predict and contribute to the pathophysiology of GvHD, although specifically in cGvHD [85]. Therefore, more studies are needed to obtain more characterized, stable, pure, and reliable human Breg populations relevant to aGvHD before foreseeing their clinical use.

### Neutrophils

Neutrophil granulocytes have been positively associated with aGvHD (especially GI-aGvHD) (Figure 1), likely through recruitment and activation of alloreactive T cells

[86–89], possibly, at least in part, in a β2-defensin–dependent manner [88] and/or microbial surface polysaccharide poly-N-acetylglucosamine [86,87]. Neutrophils recruited upon translocation of intestinal bacteria enhanced GvHD via tissue damage [90]. Moreover, gut microbiota at neutrophil engraftment was proposed as a predictive biomarker for severe aGvHD. In 141 allo-HSCT AML patients with febrile neutropenia, levels of *Lachnospiraceae*, *Peptostreptococcaceae*, *Erysipelotrichaceae*, and *Enterobacteriaceae* at day 15 post-transplantation could predict aGVHD development and correlated with a higher Treg/Th17 ratio [91]. Another study in 66 heterogeneous allo-HSCT patients showed that not only aGVHD was associated with a decreased microbiota diversity, but also, at neutrophil recovery, the presence of *Actinobacteria* or *Firmicutes* was associated with later aGVHD, whereas *Lachnospiraceae* was negatively correlated [86,92]. Overall, these studies support a significant role of neutrophils in the pathogenesis of aGVHD, although few studies are available, likely due to their short lifespan and their inability to be cryopreserved, cultured, and expanded *in vitro*.

### *Other immune cells*

Several other cell types have been reported to participate in protecting against aGvHD, especially GI-aGvHD.

ILC2 and ILC3 contribute to the control of local inflammation, suppress alloreactive T cells, and promote the maintenance and repair of the intestinal barrier via IL-22 production (Figure 1). Accordingly, elevated numbers of ILC3 in allo-HSCT grafts as well as in the patients' blood and tissues appear to protect against mucositis and aGvHD (Figure 1), strongly supporting their protective role against aGvHD [93,94]. MAIT cells, specifically recognizing microbial riboflavin derivatives presented on MR1, also exhibit regulatory and tissue repair functions [95]. They have been linked to a gut microbiota with high diversity early after allo-HSCT and with decreased GvHD incidence and increased OS [96] (Figure 1). Tissue-resident invariant NKT cells are activated by self- or bacterial-derived (e.g. *Bacteroides fragilis*) lipids presented on CD1 molecules, and commensal composition can have profound effects on them [97]. Additionally, prophylactic activation of iNKT following allo-HSCT was proposed to limit aGvHD rates [98]. NK cells can either contribute to GvHD-related inflammation or exert inhibitory functions through their KIR expression in a homeostatic environment. Studies exploring the role of these cell subsets, and other IELs, such as $CD4^+/CD8\alpha\beta^+/TCR\alpha\beta^+$, $CD4^-/CD8\alpha\beta^-/TCR\alpha\beta^+$, $CD8\alpha\alpha^+/TCR\alpha\beta^+$, and γδ T cells, on aGvHD and their interplay with gut microbiota remain scarce and hence deserve further investigations.

## Orchestration of the intricate crosstalk between gut microbiota and immune cells

The above-discussed data highlight the huge complexity of (1) gut microbiota composition, further varying between intestinal segments (duodenum, jejunum, ileum, cecum, colon, and rectum) and dynamics, and (2) heterogeneous plastic immune cell responses, especially in the gut. Unraveling their intricate crosstalk is far from being fully achieved. Numerous drawbacks, including the use of fecal microbiota in most studies instead of the actual whole mucosal microbiota ecosystem for evident practical purposes, impede such a goal. Of note, numerous studies demonstrated that, while gut microbiota composition remains relatively stable at the phylum level, significant spatiotemporal variations exist at the species or genus level, underscoring the need to understand how immune cells migrate and reside at the right place (in this case, at distinct intestinal segments/niches) and time in a coordinated and calibrated manner [99]. Another major obstacle is that different strains of the same gut microbiota species can induce markedly variable phenotypic and functional outcomes. Furthermore, tackling the high-level convolutional intrinsic network of gut bacteria interactions, for example, through cross-feeding, competitive exclusion mechanisms, or their interplay with other microbiota members (viruses, fungi, protozoa, and archaea) is required to comprehend their roles in steady-state and inflammatory conditions, which was exemplifying by a recent study revealing an association of *Lachnospiraceae*- and *Oscillospiraceae*-associated bacteriophages with immune-modulatory metabolite production [100]. Similarly, a phage-derived antibacterial enzyme was shown to lyse the *Enterococcus faecalis* biofilm, a aGvHD risk factor, both *in vitro* and *in vivo* [101]. Advancing towards a better understanding of these overall considerations is needed to design effective therapeutic strategies against aGvHD and numerous other inflammatory conditions.

## Future directions/opinion

Distinctive relationships between gut microbiota components and their ability to fine-tune host immune responses are being identified at a remarkable rate. However, the 'gut microbiota/immunoregulatory cells' intricate interplay in homeostasis and specific diseases is yet to be comprehended at a higher level of complexity. Nonetheless, there is an ever-growing body of evidence strongly supporting the instrumental roles of this crosstalk in diverse clinical settings to predict and treat most diseases. In the context of aGvHD, various pivotal factors have been identified. As detailed above, individual gut commensals (e.g. *Blautia*, *Bacteroides fragilis, Faecalibacterium duncaniae*) clearly drive immunoregulatory responses proficiently curbing detrimental inflammation in *in vitro* and preclinical studies, emphasizing the therapeutic potential for strategies targeting such components to mitigate inflammation. They could be transiently beneficial, but likely poised to fail in chronic inflammatory conditions, in which long-lasting modifications would likely be required. In the relatively short time frame between allo-HSCT and aGvHD onset, such interventions could represent substantial clinical tools. However, several hurdles persist, such as patient variability and the colossal functional heterogeneity between bacterial strains, in some cases virtually patient-specific, hence limiting the possibility to clinically target individual strains in a controlled manner. However, while underlying mechanisms remain ill-defined, microbiota diversity was clearly demonstrated to protect against aGvHD. Biotech companies have embraced this well-demonstrated fact and have been testing highly diverse microbiota products in clinical trials with very encouraging results [35].

On the other hand, other approaches bypassing initial microbiota requirement, such as cellular therapies targeting potent regulatory subsets, represent cumbersome yet manageable and promising avenues to prevent aGvHD [65]. This is epitomized, in our opinion, by DP8α Tregs, whose frequency is associated with aGvHD protection [65], and which respond to microbiota-derived and allogeneic antigens, and hence represent an invaluable target. For that matter, GMP-compliant DP8α Treg production is currently being designed.

Probiotics and prebiotics, as well as diet considerations, are expected to bring some level of protection, especially when used in combination with microbiota transfer or cellular-based approaches to further promote intestinal integrity and counteract aGvHD-related inflammation.

Importantly, such therapeutic approaches designed to curb GvHD-related inflammation should also aim at preserving the antitumoral GvL effect to limit disease relapse. While some conflicting results exist, in most preclinical and early clinical settings, Treg infusion/induction/activation can limit GvHD, while maintaining or even promoting antitumor responses (e.g. through intrinsic cytolytic activity of certain regulatory subsets (including DP8α Tregs, Tr1 cells, CD8$^+$ Tregs) or by the ability of FMT products or bacterial strains such as *F. duncaniae* or *Akkermansia muciniphila* to overcome resistance to immune checkpoint blockade) [65,102–104]. As stated above regarding anti-aGvHD responses, and undoubtedly deserving further investigation, eliciting suppressive responses at the right time and place will be critical to maintain or increase GvL responses, while precluding GvHD.

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## Author contributions

**Emmanuelle Godefroy:** Conceptualization, design, Writing – original draft, Writing – review & editing, Visualization, Project administration. **Frédéric Altare:** Writing – review & editing.

## Declaration of Competing Interest

The authors have nothing to declare.

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